Patterns, sequences and ratio

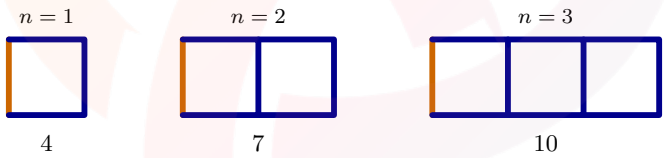
L1-AL2 Level 1 Foundations · Algebra

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Rules for sequences, square and triangular numbers, adding 1 + 2 + ... + n by pairing, and ratio and percentage problems.
By the end you can find and use a position-to-term rule for a sequence that goes up in equal steps; add 1 + 2 + ... + n and other evenly spaced numbers by pairing; solve ratio and percentage problems, including ones where an amount changes.
Prerequisites: Thinking with algebra (L1-AL1). **Contests:** JMC, Junior Kangaroo, JMO, AMC 8.
Time: about 60 minutes.

1 Lesson

Squares are made in a row from matchsticks, as shown. One square needs 4 matchsticks, two squares need 7 and three squares need 10. How many matchsticks are needed for 50 squares?



Each new square adds 3 matchsticks, so you could keep adding 3 until you reach the fiftieth pattern, but that takes 49 additions. The picture gives a quicker rule. Every pattern starts with one matchstick on the left, and each square then adds three more. So n squares need $3n + 1$ matchsticks, and 50 squares need 151. Check the rule on the patterns we can see, $3 \times 1 + 1 = 4$, $3 \times 2 + 1 = 7$ and $3 \times 3 + 1 = 10$.

1.1 Two kinds of rule

A *term-to-term* rule tells you how to get each term from the previous term, such as "add 3". A *position-to-term* rule tells you the term in position n directly, such as "$3n + 1$". The second kind is the useful one for questions about the 50th or the 1000th term.

A sequence that goes up (or down) by the same amount each time is called an *arithmetic sequence*. The amount is the common difference.

Arithmetic sequences.
If the first term is a and the common difference is d, then the nth term is $a + (n - 1)d$. The number of terms in the list $a, a + d, a + 2d, \ldots, b$ is $\frac{b-a}{d} + 1$.

To get from the first term to the nth term you take $n - 1$ steps, not n, because a row of n fence posts has only $n - 1$ gaps between them. For the same reason, the list $3, 8, 13, \ldots, 98$ has $\frac{98-3}{5} = 19$ steps and so 20 terms.

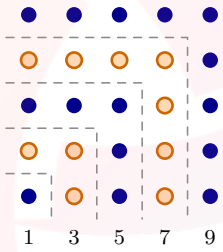
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Example 1.
The 5th term of an arithmetic sequence is 23 and the 12th term is 51. Find the 100th term.

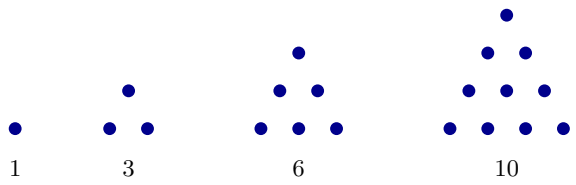
Solution.
We do not know the first term or the common difference, so we find the difference first. From the 5th term to the 12th term there are $12 - 5 = 7$ steps, and the sequence goes up by $51 - 23 = 28$. So each step is $28 \div 7 = 4$.
From the 5th term to the 100th term there are 95 steps, so the 100th term is $23 + 95 \times 4 = 403$.
(Equivalently, the first term is $23 - 4 \times 4 = 7$, and the 100th term is $7 + 99 \times 4 = 403$.)
The most common slip is to count 8 steps from the 5th term to the 12th, by counting the terms instead of the gaps.

1.2 Square and triangular numbers

The square numbers $1, 4, 9, 16, 25, \ldots$ count dots in square arrays, and the nth one is n^2. They go up by $3, 5, 7, 9, \ldots$, and the picture shows why. To turn an $(n-1)$ by $(n-1)$ square into an n by n square you add an L-shape of n dots along the top and n down the side, sharing a corner, which is $2n - 1$ dots. Building the n by n square one L-shape at a time, starting from a single dot, adds $1, 3, 5, \ldots, 2n-1$ dots. So $1 + 3 + 5 + \cdots + (2n-1) = n^2$, so the sum of the first n odd numbers is n^2.



The *triangular numbers* $1, 3, 6, 10, 15, \ldots$ count dots in triangles. To make the next triangle you add a new bottom row with one more dot than before, so the nth triangular number is $1 + 2 + 3 + \cdots + n$.



Theorem.
For every positive whole number n,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

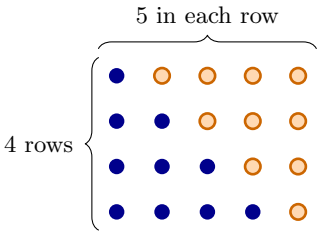
Proof. Call the sum S, and write the sum forwards and then backwards underneath.

$$S = 1 + 2 + 3 + \cdots + n,$$
$$S = n + (n-1) + (n-2) + \cdots + 1.$$

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Add the two lines column by column. Each column adds up to $n+1$, because the top number goes up by 1 from one column to the next while the bottom number goes down by 1. There are n columns, so $2S = n(n+1)$, and $S = \frac{n(n+1)}{2}$. □

The dot picture shows the same proof. Two copies of a triangle with 4 rows fit together into a rectangle of 4 rows of 5, so the triangle has half of 4×5 dots.



Pairing works for any arithmetic sequence. The first and last terms, the second and second last, and so on all have the same total. So the sum of an arithmetic sequence is the number of terms times $\frac{\text{first}+\text{last}}{2}$, and the mean of the terms is $\frac{\text{first}+\text{last}}{2}$. For example, $3+8+13+\cdots+98$ has 20 terms, so its sum is $20 \times \frac{3+98}{2} = 1010$.

Example 2.
A shop stacks 300 tins in a triangle, with one tin in the top row and each row having one more tin than the row above. How many tins are in the bottom row?

Solution.
If there are n rows, the bottom row has n tins and the total is $1+2+\cdots+n = \frac{n(n+1)}{2}$. So $\frac{n(n+1)}{2} = 300$, which means $n(n+1) = 600$.
We need two consecutive whole numbers that multiply to 600. Their product is a little more than $n \times n$, so n is a little less than the number whose square is 600. Since $24^2 = 576$ and $25^2 = 625$, try $n = 24$, which gives $24 \times 25 = 600$. The bottom row has 24 tins.

1.3 Ratio and proportion

A ratio $2 : 7$ means that for every 2 of the first thing there are 7 of the second. The easiest way to think is in *parts*. To share £45 in the ratio $2 : 7$, count $2+7 = 9$ parts, so one part is £5, and the shares are £10 and £35. Note that the first share is $\frac{2}{9}$ of the total, not $\frac{2}{7}$.

Two quantities are in proportion if multiplying one by a number multiplies the other by the same number. In that case, find the value of one unit first. If 6 pens cost £4.50, one pen costs 75p, so 10 pens cost £7.50.

When one quantity in a ratio changes and another stays the same, rewrite both ratios so that the unchanged quantity has the same number of parts in each.

Example 3.
A bag contains red and blue counters in the ratio $3 : 5$. After 12 more red counters are put in, the ratio of red to blue is $9 : 11$. How many counters were in the bag at the start?

Solution.
The number of blue counters does not change, so write both ratios with the same number of blue parts. The smallest number that both 5 and 11 divide is 55, so

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$$3 : 5 = 33 : 55 \qquad \text{and} \qquad 9 : 11 = 45 : 55.$$

The blue counters are 55 parts in both. The red counters went from 33 parts to 45 parts, an increase of 12 parts, and this was 12 counters. So one part is one counter.
At the start there were 33 red and 55 blue counters, which is 88 counters. As a check, after adding 12 red there are 45 red and 55 blue, and $45 : 55 = 9 : 11$.
With algebra, say there were $3k$ red and $5k$ blue counters at the start. Then $\frac{3k+12}{5k} = \frac{9}{11}$, and multiplying both sides by $5k$ and by 11 gives $11(3k + 12) = 9 \times 5k$, that is $33k + 132 = 45k$. So $12k = 132$, $k = 11$ and there were $8k = 88$ counters.

1.4 Percentages

Per cent means per hundred, so x% of an amount is $\frac{x}{100}$ of that amount. For changes, use a multiplier. Increasing by 20% multiplies by 1.2, because the new amount is $100\% + 20\% = 120\%$ of the old one, and decreasing by 20% multiplies by 0.8. Two changes one after the other multiply together. To find the amount before a change, divide by the multiplier.

Example 4.
A shop raises the price of a coat by 20%. Later, in a sale, the shop reduces the new price by 20%. The coat now costs £96. What was the price before both changes, and by what percentage has the price changed overall?

Solution.
You might expect the two changes to cancel, so that the original price was £96. They do not, because the 20% decrease is taken from a larger amount than the 20% increase was.
The two changes multiply the price by $1.2 \times 0.8 = 0.96$. So the original price P satisfies $0.96P = 96$, which gives $P = 100$. The coat cost £100 at the start. As a check, 20% of £100 is £20, giving £120. 20% of £120 is £24, giving £96.
Overall the price fell from £100 to £96, a decrease of 4%. Note that adding 20% back on to £96 gives £115.20, not the original price, because 20% of £96 is a different amount from 20% of £120.

Common traps.

- Counting terms instead of steps. From the 5th term to the 12th term there are 7 steps.
- Reading a ratio 2 : 7 as the fraction $\frac{2}{7}$ of the total. The correct fraction is $\frac{2}{9}$.
- Adding percentage changes. A rise of 20% and then a fall of 20% is a fall of 4%.
- Undoing a percentage change by taking the same percentage of the new amount.

1.5 How to spot this in a contest

1. For a pattern of shapes or a list that goes up in equal steps, find the position-to-term rule $a + (n - 1)d$ and check the rule on the first few terms.
2. For a long sum of evenly spaced numbers, such as $1 + 2 + \cdots + 100$ or the even numbers up to 200, pair the first with the last and use the number of terms.
3. For dots in a triangle, or rows that each have one more object than the row before, think of triangular numbers.

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4. For a ratio that changes when something is added or taken away, find the quantity that stays the same and give that quantity the same number of parts.
5. For several percentage changes one after another, multiply the multipliers.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ The sequence $6, 13, 20, 27, \ldots$ goes up by 7 each time. What is the first term of the sequence that is greater than 500?

Problem 2 ⋆ Ava and Ben share some sweets in the ratio $4 : 7$. Ben gets 18 more sweets than Ava. How many sweets are there altogether?

Problem 3 ⋆⋆ Find the value of $21 + 22 + 23 + \cdots + 80$.

Problem 4 ⋆⋆ In a class of 30 pupils, 40% are boys. Some new boys join the class, and no one leaves. Now 50% of the class are boys. How many new boys joined?

Problem 5 ⋆⋆⋆ Explain why the sum of two consecutive triangular numbers is always a square number. (For example, $6 + 10 = 16$ and $10 + 15 = 25$.)

Problem 6 ⋆⋆⋆ Lena writes down the whole numbers $1, 2, 3, \ldots, n$ for some n. She then rubs out one of them. The numbers that are left add up to 400. Which number did she rub out?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2019 Question 3** ⋆
   Compare the remaining part with the eaten part, not with the whole
2. **AMC 8 2017 Problem 2** ⋆
   You know what percentage of the total Brenda's votes are
3. **UKMT JMC 2017 Question 12** ⋆
   The fraction of pupils who are girls stays the same
4. **AMC 8 2015 Problem 9** ⋆
   The daily sales are the odd numbers
5. **UKMT JMC 2017 Question 17** ⋆⋆
   Pair each even number with an odd number next to that even number
6. **AMC 8 2020 Problem 15** ⋆⋆
   Write both percentages as fractions and compare

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7. **UKMT Junior Kangaroo 2017 Question 9** ⋆⋆
   Count the squares on each diagonal, and ask whether the diagonals share one

8. **AMC 8 2015 Problem 13** ⋆⋆
   Find the sum of the whole set first, then the sum that is left

9. **AMC 8 2016 Problem 19** ⋆⋆
   The mean of evenly spaced numbers is the middle one

10. **UKMT Junior Kangaroo 2019 Question 24** ⋆⋆⋆
    The frogs in each pond must be a whole number of ratio parts

11. **AMC 8 2024 Problem 21** ⋆⋆⋆
    The total number of frogs stays the same

12. **AMC 8 2019 Problem 22** ⋆⋆⋆
    Write each change as a multiplier

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4 Hints

1. Find a rule for the nth term first.
2. How many more parts than Ava does Ben get?
3. How many terms are there? Pair the first with the last.
4. Which number does not change when the new boys join?
5. Draw a square of dots and split the square along a staircase. Or use the formula $\frac{n(n+1)}{2}$ for both triangular numbers.
6. The full list adds up to a triangular number, and the number rubbed out is between 1 and n. Which triangular numbers are just above 400?

5 Solutions

Problem 1

The sequence $6, 13, 20, 27, \ldots$ *goes up by* 7 *each time. What is the first term of the sequence that is greater than* 500*?*

The first term is 6 and each step adds 7, so the nth term is $6 + 7(n-1) = 7n - 1$. We need $7n - 1$ to be greater than 500, that is, $7n$ greater than 501. Since $7 \times 71 = 497$ is too small and $7 \times 72 = 504$ is large enough, the first such term is the 72nd, which is $504 - 1 = 503$. As a check, the 71st term is 496, which is less than 500.

Answer: 503

Problem 2

Ava and Ben share some sweets in the ratio 4 : 7*. Ben gets* 18 *more sweets than Ava. How many sweets are there altogether?*

Ava gets 4 parts and Ben gets 7 parts, so Ben gets 3 parts more than Ava. These 3 parts are 18 sweets, so one part is 6 sweets. Altogether there are $4 + 7 = 11$ parts, which is $11 \times 6 = 66$ sweets. As a check, Ava gets 24 and Ben gets 42, which is 18 more.

Answer: 66

Problem 3

Find the value of $21 + 22 + 23 + \cdots + 80$*.*

The numbers from 21 to 80 form an arithmetic sequence with $80 - 21 + 1 = 60$ terms. Pair the first with the last, the second with the second last, and so on, so $21 + 80 = 101$, $22 + 79 = 101$, and every pair adds up to 101. The 60 terms make 30 pairs, so the sum is $30 \times 101 = 3030$.

Another way is to take $1 + 2 + \cdots + 80$ minus $1 + 2 + \cdots + 20$, which is $\frac{80 \times 81}{2} - \frac{20 \times 21}{2} = 3240 - 210 = 3030$.

Answer: 3030

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Problem 4

In a class of 30 pupils, 40% are boys. Some new boys join the class, and no one leaves. Now 50% of the class are boys. How many new boys joined?

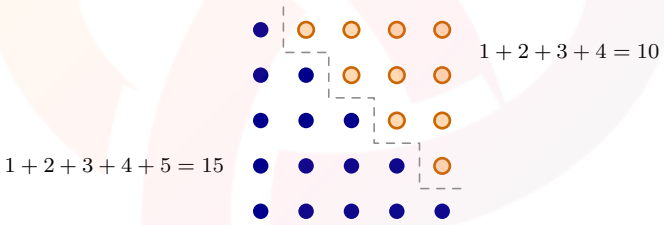
At the start there are 40% of 30, which is 12 boys, and so 18 girls. The number of girls does not change. Afterwards half the class are boys, so the other half are girls, and the class has $2 \times 18 = 36$ pupils. The class had 30 before, so 6 new boys joined. As a check, there are now 18 boys out of 36, which is 50%.

Answer: 6

Problem 5

Explain why the sum of two consecutive triangular numbers is always a square number. (For example, $6 + 10 = 16$ and $10 + 15 = 25$.)

Take an n by n square of dots and split the square by a staircase, as in the picture for $n = 5$. The dots on and below the staircase form rows of $1, 2, 3, \ldots, n$ dots, which is the nth triangular number. The dots above the staircase form rows of $0, 1, 2, \ldots, n-1$ dots, which is the triangular number before the nth. So the two consecutive triangular numbers $1 + 2 + \cdots + (n-1)$ and $1 + 2 + \cdots + n$ together make n^2.



The formula gives the same result. The two triangular numbers are $\frac{(n-1)n}{2}$ and $\frac{n(n+1)}{2}$. Their sum is

$$\frac{n(n-1) + n(n+1)}{2} = \frac{n\big((n-1) + (n+1)\big)}{2} = \frac{n \times 2n}{2} = n^2.$$

This holds for every n, so the sum of two consecutive triangular numbers is always a square.

Problem 6

Lena writes down the whole numbers $1, 2, 3, \ldots, n$ for some n. She then rubs out one of them. The numbers that are left add up to 400. Which number did she rub out?

The full list adds up to $T = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. If she rubbed out x, then $T - x = 400$, so $T = 400 + x$. Since x is one of the numbers $1, 2, \ldots, n$, the total T is at least 401 and at most $400 + n$.

The triangular numbers near 400 are $\frac{27 \times 28}{2} = 378$, $\frac{28 \times 29}{2} = 406$ and $\frac{29 \times 30}{2} = 435$. The total must be at least 401, so n is at least 28.

Now, T at most $400 + n$ means $T - n$ is at most 400. But $T - n = 1 + 2 + \cdots + (n-1)$ is the triangular number before T. For $n = 28$ this number is 378, which is fine, but for $n = 29$ the number would be 406, which is too big, and the number only grows for larger n. So $n = 28$, $T = 406$ and $x = 406 - 400 = 6$.

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As a check, $1 + 2 + \cdots + 28 = 406$, and removing 6 leaves 400. She rubbed out 6.

Answer: 6